

M-HEC User Manual

MLFF-Enhanced High-Temperature Elastic Constants

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1. Introduction and theory

M-HEC computes single-crystal elastic constants at finite temperature by combining VASP *ab initio* molecular dynamics with on-the-fly machine-learning force fields and symmetry-adapted stress-strain sampling. The elastic tensor is obtained from the stress response to finite strains through the generalised Hooke law

$$\sigma_i = C_{ij} \epsilon_j, \quad i, j = 1 \text{ to } 6 \text{ (Voigt notation)}$$

Here i and j are Voigt indices from 1 to 6 that label the stress and strain components. For each temperature the equilibrium lattice is first obtained from an NPT simulation, deformations are then sampled under fixed-cell NVT conditions, and the time-averaged stresses are regressed against the applied strain.

1.1 Strain-domain consistency

The central design principle is that the force field is trained and validated over the same strain range, and at the same thermal-expansion-equilibrium lattice, that are later used for the elastic measurement. The elastic constants are then obtained by interpolation inside the training domain rather than by extrapolation outside it. Accuracy in the strained and stressed configurations that determine the elastic constants is further preserved by an elevated stress weight during training, odd-polynomial debiasing of the fit, and a residual-pressure correction.

Scope. Strain-domain consistency guarantees interpolation rather than extrapolation. It does not by itself guarantee accuracy, which remains contingent on the quality of the underlying first-principles reference.

1.2 Capabilities

- Automatic identification of the crystal system and Laue class, with optional space-group input. All seven crystal systems, nine Laue classes, are supported.
- Two complementary deformation schemes, SC-SS and SA-SS.
- A single symmetry-parameterised, R-squared-weighted joint least-squares solver shared by both schemes, returning every independent constant with a fit standard error.
- Robust per-channel slope fitting with automatic model selection.
- Wallace residual-pressure diagnostics and the Birch to Brugger conversion.
- Born mechanical-stability checking.
- MLFF training and energy, force and stress parity validation.
- Post-processing of mechanical properties and of AIMD trajectories.
- Equilibrium lattice and thermal-expansion analysis.

2. Installation

Component	Requirement
Python	3.8 or newer
numpy	required
VASP	6.x recommended, for AIMD and on-the-fly MLFF
matplotlib	optional, diagnostic figures
spglib	optional but recommended, reliable Laue-class detection
pyyaml	optional, YAML configuration files

2.1 Installing from the source archive

Unpack the distributed tarball, enter the extracted directory and install from it. This is the recommended route.

```
tar -xzf mhec-1.0.0.tar.gz
cd mhec-1.0.0
pip install .
```

To include the optional dependencies, replace the last command with the following.

```
pip install ".[all]"
```

If the account has no write access to the system site-packages, install into the user tree instead.

```
pip install --user .
```

2.2 Alternative routes

Installation from the wheel, which requires no unpacking.

```
pip install --force-reinstall --no-deps mhec-1.0.0-py3-none-any.whl
```

Installation directly from the archive, which pip can read without unpacking.

```
pip install mhec-1.0.0.tar.gz
```

2.3 Verifying the installation

```
mhec --version
```

If the `mhec` command is not found after a user installation, add the user script directory to the search path.

```
export PATH="$HOME/.local/bin:$PATH"
```

Upgrading. When reinstalling over an existing copy, add `--force-reinstall`. Adding `--no-deps` as well keeps the already satisfied dependencies untouched, which is convenient on clusters with restricted network access.

3. Input files and directory layout

Prepare a working directory containing the following files before launching the program.

File	Role
POSCAR	structure of the reference cell, required
POTCAR	pseudopotentials, copied into every generated directory
KPOINTS	k-point mesh
INCAR	template, its key settings are inherited and adapted
run.sh	cluster submission template, its resource and environment lines are reused
ML_FF	optional pre-trained force field, enables run-only mode

Environment. The submission template should load its own environment, for example `source ~/.bashrc` followed by `conda activate` or `module load`. Scripts that rely on inheriting an interactive shell environment may fail when the job is started by a batch driver on a different node.

4. The command-line interface

Launch the interactive interface with

```
mhec
```

The header reports the version, the detected system, the temperatures and the strain settings. The numbered menu is grouped into four blocks, Elastic Constants, Post-processing, Machine-Learning Force Field and Utilities. Item `L` switches the interface language and item `0` exits.

A typical session proceeds as follows.

1. Item 10, confirm the crystal system and lattice parameters.
2. Item 7 or 8, train a force field, then item 9 to validate its accuracy.
3. Item 12, obtain the equilibrium lattice at each temperature.
4. Item 1 or 3, generate the elastic-constant inputs, then submit them.
5. Item 2 or 4, fit the elastic constants.
6. Item 5, post-process the mechanical properties.

5. Deformation schemes

5.1 SC-SS, Single-Component Stress-Strain

Each deformation activates a single Voigt strain component, so one deformation probes a complete column of the stiffness matrix. The scheme is general and applies to every crystal system, and it is the natural choice for low-symmetry crystals where many independent constants must be covered.

5.2 SA-SS, Symmetry-Adapted Stress-Strain

Deformations are matched to the symmetry of the crystal so as to maximise the measured stress signal. Volume-conserving shears are used for the shear block, and deviatoric strains are used for combinations such as C11 minus C12, which is more accurate than taking the difference of two large uniaxial numbers. For hexagonal, tetragonal and trigonal systems an `ortho_ac` mode with strain (δ , a , minus δ), where a equals δ squared divided by one minus δ squared, measures C11 plus C33 minus twice C13 through the strong differential signal of the first minus the third normal stress, which places a tight constraint on C13.

Representative relations between the measured stress slope and the elastic constants are collected below.

Mode	Measured combination	Relation
uniaxial	single normal strain	$\sigma_i' = C_{ij}$
deviatoric ab	C11 minus C12	$C11 \text{ minus } C12 = \sigma_1'$
<code>ortho_ac</code>	C11 plus C33 minus 2 C13	$= \sigma_1' \text{ minus } \sigma_3'$
volume-conserving shear	C44	$\sigma_4' = 2 C44$, hence $C44 = \sigma_4' \text{ divided by } 2$
single-component shear	C44, low symmetry	$\sigma_4' = C44$

The factor of two for volume-conserving shear follows from the engineering strain convention, ϵ_{s_4} equals twice the tensor component ϵ_{yz} . Both schemes are processed by the same solver and write the same output files, so their results can be compared directly.

6. Menu reference

6.1 Elastic Constants

Item	Function	Notes
1	SC-SS, generate inputs	Builds one directory per deformation and amplitude, copies POTCAR, KPOINTS, adapted INCAR and the submission script, and writes a batch submission driver.
2	SC-SS, fit elastic constants	Collects the averaged stresses, fits the slopes, solves the tensor, prints the raw and symmetrized matrices, writes the report and the diagnostic figures.
3	SA-SS, generate inputs	As item 1 but using the symmetry-adapted mode set for the detected Laue class.
4	SA-SS, fit elastic constants	As item 2. The screen output, provenance report and file set are identical in format to the SC-SS route.

6.2 Post-processing

Item	Function	Notes
5	Elastic post-processing	Voigt, Reuss and Hill moduli, universal anisotropy index, direction-dependent moduli and Poisson ratio, Debye temperature and sound velocities.
6	AIMD post-processing	Analyses NPT and NVT trajectories, for example radial distribution functions, mean square displacement and structural or transport indicators.

6.3 Machine-Learning Force Field

Item	Function	Notes
7	MLFF training, isothermal	Trains at a fixed target temperature. The workflow is train, refit, run. An elevated stress weight preserves accuracy in the stressed configurations that determine the elastic constants.
8	MLFF training, heating ramp	Trains along a temperature ramp so that one force field covers the whole temperature range. Training stages are chained and a final refit is submitted automatically.
9	MLFF accuracy validation	Parity of energy, force components and stress against DFT, with RMSE and MAE. Validation should be performed on the same strain domain that is used for the elastic measurement.

6.4 Utilities

Item	Function	Notes
10	Crystal system and lattice parameters	Lattice parameters and volume, composition, molar mass and density, crystal system and Laue class, number of independent constants, and the space group when spglib is available.
11	Deformation modes and strain schemes	Lists the SC-SS and SA-SS mode sets for the detected crystal system, the amplitude points and the total number of calculations.
12	Equilibrium lattice and thermal expansion	Extracts the NPT equilibrium lattice at each temperature and derives the linear and volumetric thermal-expansion coefficients. Run this before the fixed-cell NVT deformations.

Recommended order. Force field, item 7 or 8, then validation, item 9, then equilibrium lattice, item 12, then deformations, items 1 to 4, then properties, item 5. The equilibrium lattice must correspond to the same temperature as the subsequent NVT sampling.

7. Fitting and the joint solver

7.1 Robust per-channel slope fitting

For every stress channel the slope at vanishing strain is determined with an automatic model selection.

1. An odd polynomial through the origin, $\sigma = a \epsilon + b \epsilon^3$, is fitted and the linear coefficient a is taken as the slope. The odd form removes the bias introduced by the leading anharmonic term and ignores the even, volumetric part by construction. At least six non-zero points are required.
2. If the odd polynomial does not improve on a linear fit through the origin, the linear form is used instead, which prevents over-fitting of the cubic term.
3. If the selected model still does not reach the quality threshold, the outermost strain points are removed successively and the slope is taken from the largest small-strain window that reaches the threshold. Such channels are labelled as narrowed.

7.2 Symmetry-parameterised weighted joint solution

Rather than assigning one deformation to one constant, all equations from all deformations and all stress components are assembled into a single overdetermined system. The stiffness matrix is parameterised by the symmetry of the crystal, so that only the independent constants remain as unknowns, and the system is solved by least squares with weights derived from the fit quality of each channel. Clean channels dominate the solution while noisy channels are suppressed, which removes the ambiguity of obtaining several inconsistent estimates for the same constant.

Uncertainties are propagated from the weighted residual covariance and are reported for every independent constant.

7.3 Reading the provenance report

The provenance file lists the strong stress-slope equations that were used, then every independent constant in the form `Cij = value +/- standard error GPa` followed by the channels that constrain it. Channels whose fit quality falls below the threshold are listed in a separate block and flagged, narrowed channels are identified, and constants that are poorly constrained are marked so that they are not adopted without further checking.

Interpreting a large error bar. A constant reported with an unusually large standard error, or one whose only channel is flagged as weak, has not been reliably measured. Typical remedies are to raise the sampling temperature so that soft modes are thermally activated, to add amplitude points and extend the NVT sampling for that channel, and to check the force-field accuracy for the corresponding deformation.

8. Pressure correction and stability

The residual hydrostatic pressure of the reference configuration and the anisotropy of the normal stresses are reported as a diagnostic. When the magnitude of the residual pressure exceeds 0.5 GPa the Birch coefficients obtained from the fit are converted to Brugger coefficients, which are the quantities to be compared with experiment. Below that threshold the correction is negligible and is reported for information only. Born mechanical stability is evaluated for the symmetrized tensor of the relevant crystal system, and any violated condition is listed explicitly.

9. Output files

File	Content
<code>elastic_constants.txt</code>	Full report, calculation parameters, raw matrix, symmetrized Birch matrix, Wallace diagnostics, Brugger matrix, Born stability, mechanical properties.
<code>elastic.txt</code>	Plain 6 by 6 matrix in GPa, the recommended values, convenient for external programs.
<code>elastic_birch.txt</code>	Uncorrected Birch matrix, written when the pressure correction has been applied.
<code>fit_solution.txt</code>	SC-SS provenance, how each constant is solved, with standard errors and channel flags.
<code>vcfit_solution.txt</code>	SA-SS provenance, identical format.
<code>fit_*.png</code> , <code>vcfit_*.png</code>	Stress-strain diagnostic plots for the strong channels, annotated with the measured combination, the slope and the fit quality.
<code>fit_r2_heatmap.png</code>	Fit-quality heat map. Symmetry-zero and weak channels are shown in grey and are excluded from the display.
<code>fit_*_GPa.dat</code>	Measured and fitted curves in GPa, one column per strong channel, ready for external plotting.

10. Configuration and batch mode

```
# batch mode
mhec --poscar POSCAR --temp 1000 --strain 0.005 --batch

# with a configuration file
mhec --config mhec.yaml --batch --poscar POSCAR
```

Option	Description
<code>--poscar</code>	path to the POSCAR file
<code>--primitive</code>	primitive-cell POSCAR, used for crystal-system detection
<code>--crystal-system</code>	force the crystal system, overriding automatic detection
<code>--temp</code>	temperature or list of temperatures in K
<code>--strain</code>	base strain amplitude delta
<code>--method</code>	advanced strain scheme, standard or ulics
<code>--batch</code>	non-interactive batch mode
<code>--config</code>	path to a YAML or JSON configuration file
<code>--version</code>	print the version and exit

```
# mhec.yaml
temperatures: [300, 600, 900, 1200, 1500]
magnitude: 0.005          # base amplitude delta
n_points: 9              # 3, 5, 7, 9 or 11
elastic_method: sa       # sa = SA-SS, sc = SC-SS
fit_poly_order: 3        # 3 = odd polynomial, 1 = linear
space_group: 92          # optional, most reliable route to the Laue class
crystal_tol: 0.01
skip_steps: 500          # MD steps discarded before averaging
vasp:
  encut: 600
  nsw_train: 2000
  nsw_run: 10000
  potim: 1.0
```

Strain sampling may be given in three equivalent ways, with the priority `strain_points`, then `max_strain` together with `n_points`, then `magnitude` together with `n_points`. With `magnitude` equal to 0.005 and `n_points` equal to 9 the non-zero strains are plus and minus 0.005, 0.010, 0.015 and 0.020. Default point counts follow the symmetry, five for cubic and hexagonal, seven for tetragonal, trigonal and orthorhombic, and nine for monoclinic and triclinic.

11. Worked example, diamond

The distribution contains a diamond test case under `examples/diamond` with three stages, `mlff_train`, `eval_mlff` and `elastic`. Following the recommended order gives, for the frozen force field, parity errors of the order of 0.0002 eV per atom for the energy, 0.05 eV per angstrom for the force components and 1 kbar for the stress, and it reproduces the measured elastic constants of diamond and their temperature dependence over the range 300 to 1500 K. The example also illustrates the comparison of the SC-SS and SA-SS routes on the same data set.

12. Troubleshooting

Symptom	Likely cause and remedy
A constant carries a very large error bar	Its only channel is weak. Raise the temperature so that soft modes are activated, add amplitude points, extend the NVT sampling, and check the force field for that deformation.
Poor or negative fit quality for one channel	The strain amplitude exceeds the linear range, or the response is strongly anharmonic. Reduce delta to about 0.005 to 0.01. The automatic small-strain window will also narrow the fit and label the channel.
A constant that should vanish is non-zero	Insufficient sampling or an incorrect Laue class. Provide the space group in the configuration file, which is the most reliable route.
Born stability is not satisfied	Check the equilibrium lattice for the temperature in question, verify that no phase transition has occurred, and confirm that the residual pressure is small.
Elastic constants are far too stiff for a framework or multi-atom-basis crystal	Static deformations were not allowed to relax the internal ions, so the clamped-ion tensor was obtained. Relax the ions at the fixed deformed cell, that is move the ions only and keep the imposed strain.
The crystal system is detected as triclinic for an MD snapshot	Thermal displacements break the apparent symmetry. Use a time-averaged structure with a looser tolerance, or specify the space group explicitly.
Jobs fail with a missing shared library	The submission script relies on an inherited environment. Load the environment inside the script itself and make sure the library version matches the executable.

General guidance. Keep the strain amplitude inside the range used for force-field training. For soft framework materials the linear range can be narrow, so amplitudes of about 0.005 to 0.01 are preferable to 0.03. Except for monatomic Bravais lattices, static stress-strain calculations must relax the internal ions at the fixed deformed cell.